Quantium Analysis 2026

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Summary: This is an inventory analysis conducted on the vendor's potato chip stock, evaluating the performance of the potato chip inventory over a year.

In [44]:
```
# Pandas and Numpy imported for analysis
import pandas as pd
import numpy as np
```

In [45]:
```
# loaded needed dataframes for analysis
transactionData = pd.read_csv("C://Users//Nicho//Downloads//QuantiumAnalyis2026//QV
customerData = pd.read_csv("C://Users//Nicho//Downloads//QuantiumAnalyis2026//QVI_p
```

In [46]:
```
# Identify types of each column
## Note: Date needs to be altered to a date format
transactionData.dtypes
```

Out[46]:
```
DATE              int64
STORE_NBR         int64
LYLTY_CARD_NBR    int64
TXN_ID            int64
PROD_NBR          int64
PROD_NAME        object
PROD_QTY          int64
TOT_SALES       float64
dtype: object
```

In [47]:
```
# check customer data as well
customerData.dtypes # no apparent issues
```

Out[47]:
```
LYLTY_CARD_NBR     int64
LIFESTAGE         object
PREMIUM_CUSTOMER  object
dtype: object
```

In [48]:
```
# explore the dataframe
transactionData.head(10) # date seems to be in a numeric format instead of a clear
```

Out[48]:

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PROD_QTY
0	43390	1	1000	1	5	Natural Chip Compny SeaSalt175g	2
1	43599	1	1307	348	66	CCs Nacho Cheese 175g	3
2	43605	1	1343	383	61	Smiths Crinkle Cut Chips Chicken 170g	2
3	43329	2	2373	974	69	Smiths Chip Thinly S/ Cream&Onion 175g	5
4	43330	2	2426	1038	108	Kettle Tortilla ChpsHny&Jlpno Chili 150g	3
5	43604	4	4074	2982	57	Old El Paso Salsa Dip Tomato Mild 300g	1
6	43601	4	4149	3333	16	Smiths Crinkle Chips Salt & Vinegar 330g	1
7	43601	4	4196	3539	24	Grain Waves Sweet Chilli 210g	1
8	43332	5	5026	4525	42	Doritos Corn Chip Mexican Jalapeno 150g	1
9	43330	7	7150	6900	52	Grain Waves Sour Cream&Chives 210G	2

In [49]:
```
# exploring the date column to understand format
earliestdate = transactionData['DATE'].min()
latestdate = transactionData['DATE'].max()
print(f" Min: {earliestdate} \n Max: {latestdate} r") # 364 date might indicate a y
```

```
Min: 43282
Max: 43646 r
```

In [50]:
```
# identifying unique values for each column:
# Note: Date has 364 unique date Numbers and the max-min difference is 364, indicat
transactionData.nunique() # further research indicates this format is likely Excel
```

Out[50]:
```
DATE              364
STORE_NBR         272
LYLTY_CARD_NBR  72637
TXN_ID         263127
PROD_NBR          114
PROD_NAME         114
PROD_QTY            6
TOT_SALES         112
dtype: int64
```

In [51]:
```
# Creating a copy for version control
transactionData2 = transactionData.copy()
transactionData2['DATE'] = pd.to_datetime(transactionData2['DATE'], unit='D', origi
# Dates are to be interpreted as 43282 days passed since 1899-12-30 and so forth
transactionData2.head(10)
```

Out[51]:

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PROD_
0	2018-10-17	1	1000	1	5	Natural Chip Compny SeaSalt175g	
1	2019-05-14	1	1307	348	66	CCs Nacho Cheese 175g	
2	2019-05-20	1	1343	383	61	Smiths Crinkle Cut Chips Chicken 170g	
3	2018-08-17	2	2373	974	69	Smiths Chip Thinly S/ Cream&Onion 175g	
4	2018-08-18	2	2426	1038	108	Kettle Tortilla ChpsHny&Jlpno Chili 150g	
5	2019-05-19	4	4074	2982	57	Old El Paso Salsa Dip Tomato Mild 300g	
6	2019-05-16	4	4149	3333	16	Smiths Crinkle Chips Salt & Vinegar 330g	
7	2019-05-16	4	4196	3539	24	Grain Waves Sweet Chilli 210g	
8	2018-08-20	5	5026	4525	42	Doritos Corn Chip Mexican Jalapeno 150g	
9	2018-08-18	7	7150	6900	52	Grain Waves Sour Cream&Chives 210G	

In [52]:
```
transactionData2.dtypes # Data is now a date format
```

Out[52]:
```
DATE              datetime64[ns]
STORE_NBR                  int64
LYLTY_CARD_NBR             int64
TXN_ID                     int64
PROD_NBR                   int64
PROD_NAME                 object
PROD_QTY                   int64
TOT_SALES                float64
dtype: object
```

In [53]:
```
# exploring the date column under the new format
# note: the diff is not inclusive of the start date 2018-07-01 and is truly represe
# this counts 2018-07-02 to 2019-06-30 (Fencepost Error)
earliestdate = transactionData2['DATE'].min()
latestdate = transactionData2['DATE'].max()
print(f" Min: {earliestdate} \n Max: {latestdate} \n diff: {latestdate - earliestda
# with this Fencepost Error noted, we can conclude that a day is missing from the d
```

```
Min: 2018-07-01 00:00:00
Max: 2019-06-30 00:00:00
diff: 364 days 00:00:00
unique: 364
```

In [54]:
```
transactionData2.isnull().sum()
```

Out[54]:
```
DATE              0
STORE_NBR         0
LYLTY_CARD_NBR    0
TXN_ID            0
PROD_NBR          0
PROD_NAME         0
PROD_QTY          0
TOT_SALES         0
dtype: int64
```

In [55]:
```
# Identifying the missing value
accurate_range = pd.date_range(start='2018-07-01', end='2019-06-30')
missing_date = accurate_range.difference(transactionData2['DATE'])
print(f"the missing date is: {missing_date}")
```

```
the missing date is: DatetimeIndex(['2018-12-25'], dtype='datetime64[ns]', freq='D')
```

In [56]:
```
transactionData2.head()
```

Out[56]:

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PROD_
0	2018-10-17	1	1000	1	5	Natural Chip Compny SeaSalt175g	
1	2019-05-14	1	1307	348	66	CCs Nacho Cheese 175g	
2	2019-05-20	1	1343	383	61	Smiths Crinkle Cut Chips Chicken 170g	
3	2018-08-17	2	2373	974	69	Smiths Chip Thinly S/ Cream&Onion 175g	
4	2018-08-18	2	2426	1038	108	Kettle Tortilla ChpsHny&Jlpno Chili 150g	

In [57]:
```
# Ensure store numbers are representative of actual locations
print(f"Stores: {transactionData2['STORE_NBR'].nunique()} \nMaxStoreNumber: {transa
# There are 272 unique stores, and store numbers range from 1 to 272, indicating pr
```

```
Stores: 272
MaxStoreNumber: 272
MinStoreNumber: 1
```

In [58]:
```
# Produce QTY verification: numbers should be at the lowest 1
print(f"MinProductQuantity: {transactionData2['PROD_QTY'].min()} \nMaxProductQuanti
```

```
MinProductQuantity: 1
MaxProductQuantity: 200
```

In [59]:
```
# Examining product names to ensure they are correct
transactionData2['PROD_NAME'].describe()
# we are looking at 114 different products
```

Out[59]:
```
count                                 264836
unique                                   114
top       Kettle Mozzarella   Basil & Pesto 175g
freq                                    3304
Name: PROD_NAME, dtype: object
```

In [60]:
```
transactionData2['PROD_NAME'].value_counts()
# woolworths medium Salsa, indicates non-chip products may be included
```

Out[60]:

PROD_NAME	
Kettle Mozzarella	Basil & Pesto 175g 3304
Kettle Tortilla Chps	Hny&Jlpno Chili 150g 3296
Cobs Popd Swt/Chlli	&Sr/Cream Chips 110g 3269
Tyrrells Crisps	Ched & Chives 165g 3268
Cobs Popd Sea Salt	Chips 110g 3265
Kettle 135g Swt Pot	Sea Salt 3257
Tostitos Splash Of	Lime 175g 3252
Infuzions Thai Sweet	Chili PotatoMix 110g 3242
Smiths Crnkle Chip	Orgnl Big Bag 380g 3233
Thins Potato Chips	Hot & Spicy 175g 3229
Kettle Sensations	Camembert & Fig 150g 3219
Doritos Corn Chips	Cheese Supreme 170g 3217
Pringles Barbeque	134g 3210
Doritos Corn Chip Mexican	Jalapeno 150g 3204
Kettle Sweet Chilli And Sour	Cream 175g 3200
Smiths Crinkle Chips Salt &	Vinegar 330g 3197
Thins Chips Light&	Tangy 175g 3188
Dorito Corn Chp	Supreme 380g 3185
Pringles Sweet&Spcy	BBQ 134g 3177
Infuzions BBQ Rib	Prawn Crackers 110g 3174
Tyrrells Crisps	Lightly Salted 165g 3174
Kettle Sea Salt	And Vinegar 175g 3173
Doritos Corn Chip Southern	Chicken 150g 3172
Twisties Chicken270g	3170
Twisties Cheese	Burger 250g 3169
Grain Waves	Sweet Chilli 210g 3167
Pringles SourCream	Onion 134g 3162
Doritos Corn Chips	Nacho Cheese 170g 3160
Cobs Popd Sour Crm	&Chives Chips 110g 3159
Kettle Original 175g	3159
Pringles Original	Crisps 134g 3157
Cheezels Cheese 330g	3149
Kettle Honey Soy	Chicken 175g 3148
Kettle Tortilla Chps	Btroot&Ricotta 150g 3146
Tostitos Smoked	Chipotle 175g 3145
Infzns Crn Crnchers	Tangy Gcamole 110g 3144
Smiths Crinkle	Original 330g 3142
Kettle Tortilla Chps	Feta&Garlic 150g 3138
Infuzions SourCream&	Herbs Veg Strws 110g 3134
Kettle Sensations	Siracha Lime 150g 3127
Old El Paso Salsa	Dip Chnky Tom Ht300g 3125
Doritos Corn Chips	Original 170g 3121
Doritos Mexicana	170g 3115
Twisties Cheese	270g 3115
Thins Chips Seasoned	chicken 175g 3114
Old El Paso Salsa	Dip Tomato Med 300g 3114
Pringles Mystery	Flavour 134g 3114
Grain Waves Sour	Cream&Chives 210G 3105
Pringles Chicken	Salt Crips 134g 3104
Thins Chips Salt &	Vinegar 175g 3103
Pringles Slt Vingar	134g 3095
Old El Paso Salsa	Dip Tomato Mild 300g 3085
Kettle Sensations	BBQ&Maple 150g 3083
Pringles Sthrn Fried	Chicken 134g 3083
Tostitos Lightly	Salted 175g 3074

Doritos Cheese	Supreme 330g	3052
Kettle Chilli 175g		3038
Smiths Chip Thinly	Cut Original 175g	1614
Snbts Whlgrn Crisps	Cheddr&Mstrd 90g	1576
Natural Chip Co	Tmato Hrb&Spce 175g	1572
Burger Rings 220g		1564
Natural ChipCo Sea	Salt & Vinegr 175g	1550
CCs Tasty Cheese	175g	1539
RRD SR Slow Rst	Pork Belly 150g	1526
Smiths Thinly Cut	Roast Chicken 175g	1519
RRD Sweet Chilli &	Sour Cream 165g	1516
Woolworths Cheese	Rings 190g	1516
CCs Original 175g		1514
RRD Honey Soy	Chicken 165g	1513
Smith Crinkle Cut	Mac N Cheese 150g	1512
WW Supreme Cheese	Corn Chips 200g	1509
Infuzions Mango	Chutny Papadums 70g	1507
RRD Chilli&	Coconut 150g	1506
Smiths Crinkle Cut	Snag&Sauce 150g	1503
Red Rock Deli Sp	Salt & Truffle 150G	1498
CCs Nacho Cheese	175g	1498
WW Original Corn	Chips 200g	1495
Red Rock Deli Thai	Chilli&Lime 150g	1495
Woolworths Mild	Salsa 300g	1491
Smiths Crinkle Cut	Chips Barbecue 170g	1489
WW Original Stacked	Chips 160g	1487
Smiths Crinkle Cut	Chips Chicken 170g	1484
WW Sour Cream &Onion	Stacked Chips 160g	1483
Smiths Crinkle Cut	Chips Chs&Onion170g	1481
Cheetos Chs & Bacon	Balls 190g	1479
RRD Salt & Vinegar	165g	1474
RRD Lime & Pepper	165g	1473
Smiths Chip Thinly	S/Cream&Onion 175g	1473
Doritos Salsa Mild	300g	1472
Smiths Crinkle Cut	Tomato Salsa 150g	1470
WW D/Style Chip	Sea Salt 200g	1469
Natural Chip	Compny SeaSalt175g	1468
GrnWves Plus Btroot	& Chilli Jam 180g	1468
WW Crinkle Cut	Chicken 175g	1467
Smiths Thinly	Swt Chli&S/Cream175G	1461
Smiths Crinkle Cut	Chips Original 170g	1461
Natural ChipCo	Hony Soy Chckn175g	1460
Red Rock Deli SR	Salsa & Mzzrlla 150g	1458
Smiths Crinkle Cut	Salt & Vinegar 170g	1455
RRD Steak &	Chimuchurri 150g	1455
Cheezels Cheese Box	125g	1454
Smith Crinkle Cut	Bolognese 150g	1451
Doritos Salsa	Medium 300g	1449
Cheetos Puffs 165g		1448
Thins Chips	Originl saltd 175g	1441
Smiths Chip Thinly	CutSalt/Vinegr175g	1440
Smiths Crinkle Cut	French OnionDip 150g	1438
Red Rock Deli Chikn&	Garlic Aioli 150g	1434
Sunbites Whlegrn	Crisps Frch/Onin 90g	1432
RRD Pc Sea Salt	165g	1431
Woolworths Medium	Salsa 300g	1430

```
NCC Sour Cream &    Garden Chives 175g      1419
French Fries Potato Chips 175g              1418
WW Crinkle Cut      Original 175g           1410
Name: count, dtype: int64
```

In [61]:
```
# Ensuring all products are chips
# Split all unique product names into individual words
product_words = (
    transactionData2['PROD_NAME']
    .unique()                          # get unique product names
    .tolist()                          # convert to list
)

# Join all product names into one string, then split into individual words
product_words = pd.Series(
    ' '.join(product_words).split()
)

print(product_words.value_counts())
```

175g	26
Chips	21
150g	19
&	17
Smiths	16
Crinkle	14
Cut	14
Kettle	13
Salt	12
Cheese	12
Original	10
Salsa	9
Chip	9
Doritos	9
Pringles	8
134g	8
170g	8
Corn	8
165g	8
RRD	8
WW	7
Chicken	7
110g	7
300g	6
Sea	6
Sour	6
Thins	5
Vinegar	5
Chilli	5
Thinly	5
Crisps	5
Cream	4
Rock	4
Deli	4
Supreme	4
Infuzions	4
Natural	4
Red	4
330g	4
Dip	3
Old	3
Tomato	3
200g	3
Mild	3
El	3
CCs	3
Sweet	3
Twisties	3
Lime	3
Cobs	3
Sensations	3
Popd	3
Tostitos	3
Soy	3
Paso	3
Woolworths	3

Tortilla	3
ChipCo	2
And	2
Thai	2
Tyrrells	2
Lightly	2
160g	2
Swt	2
BBQ	2
SR	2
Honey	2
French	2
Cheezels	2
Salted	2
Smith	2
Nacho	2
Waves	2
Potato	2
Cheetos	2
Grain	2
380g	2
Tangy	2
90g	2
Burger	2
Rings	2
190g	2
Chives	2
Medium	2
Barbecue	1
Crips	1
Stacked	1
CutSalt/Vinegr175g	1
Aioli	1
D/Style	1
BBQ&Maple	1
Fries	1
Med	1
Pc	1
Chilli&	1
Coconut	1
Hot	1
Spicy	1
Crm	1
Belly	1
Puffs	1
&Chives	1
Crnkle	1
Orgnl	1
Big	1
Bag	1
Pork	1
Ched	1
Splash	1
Chimuchurri	1
Strws	1
ChpsFeta&Garlic	1

Mango	1
Chutny	1
Papadums	1
70g	1
Steak	1
Sunbites	1
Hrb&Spce	1
Whlegrn	1
Frch/Onin	1
Snag&Sauce	1
&OnionStacked	1
Pepper	1
Vinegr	1
Chikn&Garlic	1
Veg	1
SourCream&Herbs	1
Of	1
Snbts	1
ChpsBtroot&Ricotta	1
Tasty	1
Smoked	1
Chipotle	1
Barbeque	1
Mystery	1
Flavour	1
Whlgrn	1
Slow	1
Cheddr&Mstrd	1
Chs	1
Bacon	1
Balls	1
Rst	1
Slt	1
Vingar	1
Chs&Onion170g	1
SweetChili	1
Tmato	1
Co	1
125g	1
Infzns	1
Crn	1
Crnchers	1
Gcamole	1
Chilli&Lime	1
Sthrn	1
FriedChicken	1
Sweet&Spcy	1
Mzzrlla	1
Originl	1
saltd	1
Sp	1
Truffle	1
150G	1
Box	1
Southern	1
Garden	1

```
Mexican               1
Compny                1
SeaSalt175g           1
S/Cream&Onion         1
ChpsHny&Jlpno         1
Chili                 1
210g                  1
Jalapeno              1
NCC                   1
Cream&Chives          1
210G                  1
Siracha               1
270g                  1
Light&                1
220g                  1
Chli&S/Cream175G      1
Mexicana              1
OnionDip              1
SourCream             1
Plus                  1
Btroot                1
Jam                   1
180g                  1
135g                  1
Pot                   1
Onion                 1
Crackers              1
250g                  1
Chnky                 1
Tom                   1
Ht300g                1
Swt/Chlli             1
&Sr/Cream             1
GrnWves               1
Prawn                 1
Hony                  1
Basil                 1
Chckn175g             1
Dorito                1
Chp                   1
Chicken270g           1
Roast                 1
Mozzarella            1
Pesto                 1
Rib                   1
PotatoMix             1
Camembert             1
Fig                   1
Mac                   1
N                     1
Seasonedchicken       1
Bolognese             1
Name: count, dtype: int64
```

In [62]:
```
import re
```

```
# Remove digits
product_words = product_words[~product_words.str.contains(r'\d', regex=True)]

# Remove special characters (keep only alphabetic words)
product_words = product_words[product_words.str.match(r'^[a-zA-Z]+$')]

# View word frequency high to low
word_freq = product_words.value_counts().reset_index()
word_freq.columns = ['WORD', 'COUNT']
print(word_freq)
```

	WORD	COUNT
0	Chips	21
1	Smiths	16
2	Crinkle	14
3	Cut	14
4	Kettle	13
5	Salt	12
6	Cheese	12
7	Original	10
8	Doritos	9
9	Chip	9
10	Salsa	9
11	Corn	8
12	Pringles	8
13	RRD	8
14	Chicken	7
15	WW	7
16	Sea	6
17	Sour	6
18	Thinly	5
19	Crisps	5
20	Vinegar	5
21	Thins	5
22	Chilli	5
23	Supreme	4
24	Deli	4
25	Rock	4
26	Red	4
27	Cream	4
28	Natural	4
29	Infuzions	4
30	Mild	3
31	Old	3
32	Dip	3
33	Tostitos	3
34	Tomato	3
35	CCs	3
36	Paso	3
37	El	3
38	Tortilla	3
39	Soy	3
40	Cobs	3
41	Popd	3
42	Twisties	3
43	Lime	3
44	Woolworths	3
45	Sensations	3
46	Sweet	3
47	Swt	2
48	French	2
49	ChipCo	2
50	Tyrrells	2
51	Nacho	2
52	Smith	2
53	Honey	2
54	Cheetos	2

55	BBQ	2
56	Lightly	2
57	Salted	2
58	Medium	2
59	SR	2
60	Potato	2
61	Cheezels	2
62	Waves	2
63	Burger	2
64	Tangy	2
65	Chives	2
66	Grain	2
67	And	2
68	Thai	2
69	Rings	2
70	Barbecue	1
71	Rst	1
72	Belly	1
73	Puffs	1
74	Stacked	1
75	Crips	1
76	Bag	1
77	Pork	1
78	Splash	1
79	Chimuchurri	1
80	Of	1
81	Orgnl	1
82	Crnkle	1
83	Crm	1
84	Spicy	1
85	Pc	1
86	Hot	1
87	Coconut	1
88	Compny	1
89	Big	1
90	Slow	1
91	Tasty	1
92	Pepper	1
93	Sunbites	1
94	Papadums	1
95	Chutny	1
96	Mango	1
97	Strws	1
98	Veg	1
99	Vingar	1
100	Whlegrn	1
101	Slt	1
102	Med	1
103	Balls	1
104	Bacon	1
105	Steak	1
106	Chs	1
107	Vinegr	1
108	Whlgrn	1
109	Snbts	1
110	Ched	1

111	Flavour	1
112	Aioli	1
113	Mystery	1
114	Barbeque	1
115	Chipotle	1
116	Smoked	1
117	Mexican	1
118	Btroot	1
119	Fries	1
120	Roast	1
121	Dorito	1
122	Hony	1
123	OnionDip	1
124	Mexicana	1
125	Truffle	1
126	Sp	1
127	saltd	1
128	Originl	1
129	Mzzrlla	1
130	FriedChicken	1
131	Sthrn	1
132	Chili	1
133	Gcamole	1
134	Crnchers	1
135	Crn	1
136	Infzns	1
137	Box	1
138	Southern	1
139	Garden	1
140	NCC	1
141	Siracha	1
142	Chp	1
143	Mozzarella	1
144	Tmato	1
145	Basil	1
146	Co	1
147	Tom	1
148	Chnky	1
149	Onion	1
150	SourCream	1
151	Pot	1
152	Jam	1
153	Jalapeno	1
154	Plus	1
155	GrnWves	1
156	Crackers	1
157	Prawn	1
158	Rib	1
159	Seasonedchicken	1
160	N	1
161	Mac	1
162	Fig	1
163	Camembert	1
164	PotatoMix	1
165	SweetChili	1
166	Pesto	1

```
167        Bolognese      1
```

In [63]:
```
pd.set_option('display.max_rows', None)
word_freq
```

Out[63]:

	WORD	COUNT
0	Chips	21
1	Smiths	16
2	Crinkle	14
3	Cut	14
4	Kettle	13
5	Salt	12
6	Cheese	12
7	Original	10
8	Doritos	9
9	Chip	9
10	Salsa	9
11	Corn	8
12	Pringles	8
13	RRD	8
14	Chicken	7
15	WW	7
16	Sea	6
17	Sour	6
18	Thinly	5
19	Crisps	5
20	Vinegar	5
21	Thins	5
22	Chilli	5
23	Supreme	4
24	Deli	4
25	Rock	4
26	Red	4
27	Cream	4
28	Natural	4
29	Infuzions	4

	WORD	COUNT
30	Mild	3
31	Old	3
32	Dip	3
33	Tostitos	3
34	Tomato	3
35	CCs	3
36	Paso	3
37	El	3
38	Tortilla	3
39	Soy	3
40	Cobs	3
41	Popd	3
42	Twisties	3
43	Lime	3
44	Woolworths	3
45	Sensations	3
46	Sweet	3
47	Swt	2
48	French	2
49	ChipCo	2
50	Tyrrells	2
51	Nacho	2
52	Smith	2
53	Honey	2
54	Cheetos	2
55	BBQ	2
56	Lightly	2
57	Salted	2
58	Medium	2
59	SR	2

	WORD	COUNT
60	Potato	2
61	Cheezels	2
62	Waves	2
63	Burger	2
64	Tangy	2
65	Chives	2
66	Grain	2
67	And	2
68	Thai	2
69	Rings	2
70	Barbecue	1
71	Rst	1
72	Belly	1
73	Puffs	1
74	Stacked	1
75	Crips	1
76	Bag	1
77	Pork	1
78	Splash	1
79	Chimuchurri	1
80	Of	1
81	Orgnl	1
82	Crnkle	1
83	Crm	1
84	Spicy	1
85	Pc	1
86	Hot	1
87	Coconut	1
88	Compny	1
89	Big	1

	WORD	COUNT
90	Slow	1
91	Tasty	1
92	Pepper	1
93	Sunbites	1
94	Papadums	1
95	Chutny	1
96	Mango	1
97	Strws	1
98	Veg	1
99	Vingar	1
100	Whlegrn	1
101	Slt	1
102	Med	1
103	Balls	1
104	Bacon	1
105	Steak	1
106	Chs	1
107	Vinegr	1
108	Whlgrn	1
109	Snbts	1
110	Ched	1
111	Flavour	1
112	Aioli	1
113	Mystery	1
114	Barbeque	1
115	Chipotle	1
116	Smoked	1
117	Mexican	1
118	Btroot	1
119	Fries	1

	WORD	COUNT
120	Roast	1
121	Dorito	1
122	Hony	1
123	OnionDip	1
124	Mexicana	1
125	Truffle	1
126	Sp	1
127	saltd	1
128	Originl	1
129	Mzzrlla	1
130	FriedChicken	1
131	Sthrn	1
132	Chili	1
133	Gcamole	1
134	Crnchers	1
135	Crn	1
136	Infzns	1
137	Box	1
138	Southern	1
139	Garden	1
140	NCC	1
141	Siracha	1
142	Chp	1
143	Mozzarella	1
144	Tmato	1
145	Basil	1
146	Co	1
147	Tom	1
148	Chnky	1
149	Onion	1

	WORD	COUNT
150	SourCream	1
151	Pot	1
152	Jam	1
153	Jalapeno	1
154	Plus	1
155	GrnWves	1
156	Crackers	1
157	Prawn	1
158	Rib	1
159	Seasonedchicken	1
160	N	1
161	Mac	1
162	Fig	1
163	Camembert	1
164	PotatoMix	1
165	SweetChili	1
166	Pesto	1
167	Bolognese	1

In [64]:
```
salsas = transactionData2[transactionData2['PROD_NAME'].str.contains("Salsa") == Tr
salsas['PROD_NAME'].unique()
# All need to be removed, other than Smith's Crinkle Cut  Tomato Salsa 150g and Red
```

Out[64]:
```
array(['Old El Paso Salsa   Dip Tomato Mild 300g',
       'Red Rock Deli SR    Salsa & Mzzrlla 150g',
       'Smiths Crinkle Cut  Tomato Salsa 150g',
       'Doritos Salsa       Medium 300g',
       'Old El Paso Salsa   Dip Chnky Tom Ht300g',
       'Woolworths Mild     Salsa 300g',
       'Old El Paso Salsa   Dip Tomato Med 300g',
       'Woolworths Medium   Salsa 300g', 'Doritos Salsa Mild  300g'],
      dtype=object)
```

In [127…
```
'''
transactionData2[transactionData2['PROD_NAME'].str.contains("Mac") == True]: Checke

transactionData2[transactionData2['PROD_NAME'].str.contains("RRD") == True]: Check,

transactionData2[transactionData2['PROD_NAME'].str.contains("Chicken") == True]: Ch
```

```
Note: Red Rock Deli and RDD are the same, need to check abbreviations

Check = transactionData2[transactionData2['PROD_NAME'].str.contains("Dip") == True]
Check['PROD_NAME'].unique()

Old El Paso Salsa Dip must be removed; Smith's Crinkle Cut  French Onion Dip refers
Because I am noticing a trend with Old El Paso, I am going to isolate specified row

Old_El = transactionData2[transactionData2['PROD_NAME'].str.contains("Old") == True
Old_El['PROD_NAME'].unique()
Old El Paso is only connected to Salsa

Doritos_salsa = transactionData2[transactionData2['PROD_NAME'].str.contains("Dorito
Doritos_salsa['PROD_NAME'].unique()
Doritos = transactionData2[transactionData2['PROD_NAME'].str.contains("Doritos") ==
Doritos['PROD_NAME'].unique()

Woolworths = transactionData2[transactionData2['PROD_NAME'].str.contains("Woolworth
Woolworths['PROD_NAME'].unique()

Woolworths does not make potato chips; therefore can be removed

corn = transactionData2[transactionData2['PROD_NAME'].str.contains("Corn") == True]
corn['PROD_NAME'].unique()

All, including corn, do not relate to potato chips

WW = transactionData2[transactionData2['PROD_NAME'].str.contains("WW") == True]
WW['PROD_NAME'].unique()

WW is fine, removing Corn cleans WW rows

Strws = transactionData2[transactionData2['PROD_NAME'].str.contains("Strws") == Tru
Strws['PROD_NAME'].unique()

Straws = transactionData2[transactionData2['PROD_NAME'].str.contains("Straws") == T
Straws['PROD_NAME'].unique()

Tostitos = transactionData2[transactionData2['PROD_NAME'].str.contains("Tostitos")
Tostitos['PROD_NAME'].unique()

Cheetos = transactionData2[transactionData2['PROD_NAME'].str.contains("Cheetos") ==
Cheetos['PROD_NAME'].unique()

Bolognese = transactionData2[transactionData2['PROD_NAME'].str.contains("Bolognese"
Bolognese['PROD_NAME'].unique()
'''
Keyword_removed = ['Old', 'Doritos', 'Whlgrn','Woolworths', 'Crackers','Corn', 'Str
Cleaned_PC = transactionData2[transactionData2['PROD_NAME'].str.contains('|'.join(K
Cleaned_PC['PROD_NAME'].value_counts()
```

Out[127…

PROD_NAME		
Kettle Mozzarella	Basil & Pesto 175g	3304
Cobs Popd Swt/Chlli	&Sr/Cream Chips 110g	3269
Tyrrells Crisps	Ched & Chives 165g	3268
Cobs Popd Sea Salt	Chips 110g	3265
Kettle 135g Swt Pot	Sea Salt	3257
Infuzions Thai Sweet	Chili PotatoMix 110g	3242
Smiths Crnkle Chip	Orgnl Big Bag 380g	3233
Thins Potato Chips	Hot & Spicy 175g	3229
Kettle Sensations	Camembert & Fig 150g	3219
Pringles Barbeque	134g	3210
Kettle Sweet Chilli	And Sour Cream 175g	3200
Smiths Crinkle Chips	Salt & Vinegar 330g	3197
Thins Chips Light&	Tangy 175g	3188
Pringles Sweet&Spcy	BBQ 134g	3177
Tyrrells Crisps	Lightly Salted 165g	3174
Kettle Sea Salt	And Vinegar 175g	3173
Pringles SourCream	Onion 134g	3162
Kettle Original 175g		3159
Cobs Popd Sour Crm	&Chives Chips 110g	3159
Pringles Original	Crisps 134g	3157
Kettle Honey Soy	Chicken 175g	3148
Infzns Crn Crnchers	Tangy Gcamole 110g	3144
Smiths Crinkle	Original 330g	3142
Kettle Sensations	Siracha Lime 150g	3127
Thins Chips Seasoned	chicken 175g	3114
Pringles Mystery	Flavour 134g	3114
Pringles Chicken	Salt Crips 134g	3104
Thins Chips Salt &	Vinegar 175g	3103
Pringles Slt Vingar	134g	3095
Pringles Sthrn Fried	Chicken 134g	3083
Kettle Sensations	BBQ&Maple 150g	3083
Kettle Chilli 175g		3038
Smiths Chip Thinly	Cut Original 175g	1614
Natural Chip Co	Tmato Hrb&Spce 175g	1572
Burger Rings 220g		1564
Natural ChipCo Sea	Salt & Vinegr 175g	1550
RRD SR Slow Rst	Pork Belly 150g	1526
Smiths Thinly Cut	Roast Chicken 175g	1519
RRD Sweet Chilli &	Sour Cream 165g	1516
RRD Honey Soy	Chicken 165g	1513
Smith Crinkle Cut	Mac N Cheese 150g	1512
Infuzions Mango	Chutny Papadums 70g	1507
RRD Chilli&	Coconut 150g	1506
Smiths Crinkle Cut	Snag&Sauce 150g	1503
Red Rock Deli Sp	Salt & Truffle 150G	1498
Red Rock Deli Thai	Chilli&Lime 150g	1495
Smiths Crinkle Cut	Chips Barbecue 170g	1489
WW Original Stacked	Chips 160g	1487
Smiths Crinkle Cut	Chips Chicken 170g	1484
WW Sour Cream &Onion	Stacked Chips 160g	1483
Smiths Crinkle Cut	Chips Chs&Onion170g	1481
RRD Salt & Vinegar	165g	1474
Smiths Chip Thinly	S/Cream&Onion 175g	1473
RRD Lime & Pepper	165g	1473
Smiths Crinkle Cut	Tomato Salsa 150g	1470

```
WW D/Style Chip     Sea Salt 200g           1469
Natural Chip        Compny SeaSalt175g      1468
WW Crinkle Cut      Chicken 175g            1467
Smiths Crinkle Cut  Chips Original 170g     1461
Smiths Thinly       Swt Chli&S/Cream175G    1461
Natural ChipCo      Hony Soy Chckn175g      1460
Red Rock Deli SR    Salsa & Mzzrlla 150g    1458
Smiths Crinkle Cut  Salt & Vinegar 170g     1455
RRD Steak &         Chimuchurri 150g        1455
Smith Crinkle Cut   Bolognese 150g          1451
Thins Chips         Originl saltd 175g      1441
Smiths Chip Thinly  CutSalt/Vinegr175g      1440
Smiths Crinkle Cut  French OnionDip 150g    1438
Red Rock Deli Chikn&Garlic Aioli 150g       1434
RRD Pc Sea Salt     165g                    1431
NCC Sour Cream &    Garden Chives 175g      1419
French Fries Potato Chips 175g              1418
WW Crinkle Cut      Original 175g           1410
Name: count, dtype: int64
```

In [128…
```
Cleaned_PC.head()
```

Out[128…

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PROD_
0	2018-10-17	1	1000	1	5	Natural Chip Compny SeaSalt175g	
2	2019-05-20	1	1343	383	61	Smiths Crinkle Cut Chips Chicken 170g	
3	2018-08-17	2	2373	974	69	Smiths Chip Thinly S/ Cream&Onion 175g	
6	2019-05-16	4	4149	3333	16	Smiths Crinkle Chips Salt & Vinegar 330g	
10	2019-05-17	7	7215	7176	16	Smiths Crinkle Chips Salt & Vinegar 330g	

In [129…
```
Cleaned_PC.describe()
```

Out[129…

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR
count	162282	162282.000000	1.622820e+05	1.622820e+05	162282.000000
mean	2018-12-30 04:30:29.348911104	135.034249	1.354939e+05	1.351127e+05	55.601508
min	2018-07-01 00:00:00	1.000000	1.000000e+03	1.000000e+00	1.000000
25%	2018-09-30 00:00:00	70.000000	7.005925e+04	6.783250e+04	26.000000
50%	2018-12-30 00:00:00	130.000000	1.302610e+05	1.347315e+05	53.000000
75%	2019-03-31 00:00:00	203.000000	2.030340e+05	2.024145e+05	85.000000
max	2019-06-30 00:00:00	272.000000	2.373711e+06	2.415841e+06	114.000000
std	NaN	76.663024	8.028987e+04	7.808107e+04	34.599058

In [130…

```
# Filter Data to find outliers
# Produce QTY verification: numbers should be at the lowest 1
print(f"MinProductQuantity: {Cleaned_PC['PROD_QTY'].min()} \nMaxProductQuantity: {C
```

```
MinProductQuantity: 1
MaxProductQuantity: 5
```

In [131…

```
# Produce QTY verification: numbers should be at the lowest 1
print(f"MinProductQuantity: {transactionData2['PROD_QTY'].min()} \nMaxProductQuanti
```

```
MinProductQuantity: 1
MaxProductQuantity: 200
```

In [132…

```
transactionData2[ transactionData2['PROD_QTY'] == 200 ]
```

Out[132…

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PRC
69762	2018-08-19	226	226000	226201	4	Dorito Corn Chp Supreme 380g	
69763	2019-05-20	226	226000	226210	4	Dorito Corn Chp Supreme 380g	

In [133…

```
'''
The maximum we seen ealier were corn chips not potato, so it was removed from analy
Based on the purchase data and these two being the same amount, this might be for c
'''
```

Out[133… '\nThe maximum we seen ealier were corn chips not potato, so it was removed from analysis, however this seemed to be the only outlier.\nBased on the purchase data and these two being the same amount, this might be for commerical use.\n'

In [134…
```
Cleaned_PC.nunique()

print(f"""
Cleaned Data Summary:
---------------------
Time Period   : 1 year of data
Products      : {Cleaned_PC['PROD_NAME'].nunique()} unique chip flavors/brands
Stores        : {Cleaned_PC['STORE_NBR'].nunique()} unique stores
Customers     : {Cleaned_PC['LYLTY_CARD_NBR'].nunique()} unique customers
Transactions  : {Cleaned_PC['TXN_ID'].nunique()} unique transactions
""")
```

```
Cleaned Data Summary:
---------------------
Time Period   : 1 year of data
Products      : 73 unique chip flavors/brands
Stores        : 269 unique stores
Customers     : 62184 unique customers
Transactions  : 161657 unique transactions
```

In [135…
```
customerData.head()
```

Out[135…

	LYLTY_CARD_NBR	LIFESTAGE	PREMIUM_CUSTOMER
0	1000	YOUNG SINGLES/COUPLES	Premium
1	1002	YOUNG SINGLES/COUPLES	Mainstream
2	1003	YOUNG FAMILIES	Budget
3	1004	OLDER SINGLES/COUPLES	Mainstream
4	1005	MIDAGE SINGLES/COUPLES	Mainstream

In [158…
```
customerData.describe()
```

Out[158…

	LYLTY_CARD_NBR
count	7.263700e+04
mean	1.361859e+05
std	8.989293e+04
min	1.000000e+03
25%	6.620200e+04
50%	1.340400e+05
75%	2.033750e+05
max	2.373711e+06

In [160…
```
customerData['LIFESTAGE'].value_counts()
```

Out[160…
```
LIFESTAGE
RETIREES                  14805
OLDER SINGLES/COUPLES     14609
YOUNG SINGLES/COUPLES     14441
OLDER FAMILIES             9780
YOUNG FAMILIES             9178
MIDAGE SINGLES/COUPLES     7275
NEW FAMILIES               2549
Name: count, dtype: int64
```

In [161…
```
customerData['PREMIUM_CUSTOMER'].value_counts()
```

Out[161…
```
PREMIUM_CUSTOMER
Mainstream    29245
Budget        24470
Premium       18922
Name: count, dtype: int64
```

In [162…
```
customerData.isnull().sum()
```

Out[162…
```
LYLTY_CARD_NBR      0
LIFESTAGE           0
PREMIUM_CUSTOMER    0
dtype: int64
```

In [136…
```
concat_data = Cleaned_PC.merge(customerData, on = 'LYLTY_CARD_NBR')
```

In [137…
```
concat_data.head()
```

Out[137…

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PROD_Q
0	2018-10-17	1	1000	1	5	Natural Chip Compny SeaSalt175g	
1	2019-05-20	1	1343	383	61	Smiths Crinkle Cut Chips Chicken 170g	
2	2018-08-17	2	2373	974	69	Smiths Chip Thinly S/ Cream&Onion 175g	
3	2019-05-16	4	4149	3333	16	Smiths Crinkle Chips Salt & Vinegar 330g	
4	2019-05-17	7	7215	7176	16	Smiths Crinkle Chips Salt & Vinegar 330g	

In [138…
```
concat_data.isnull().sum()
```

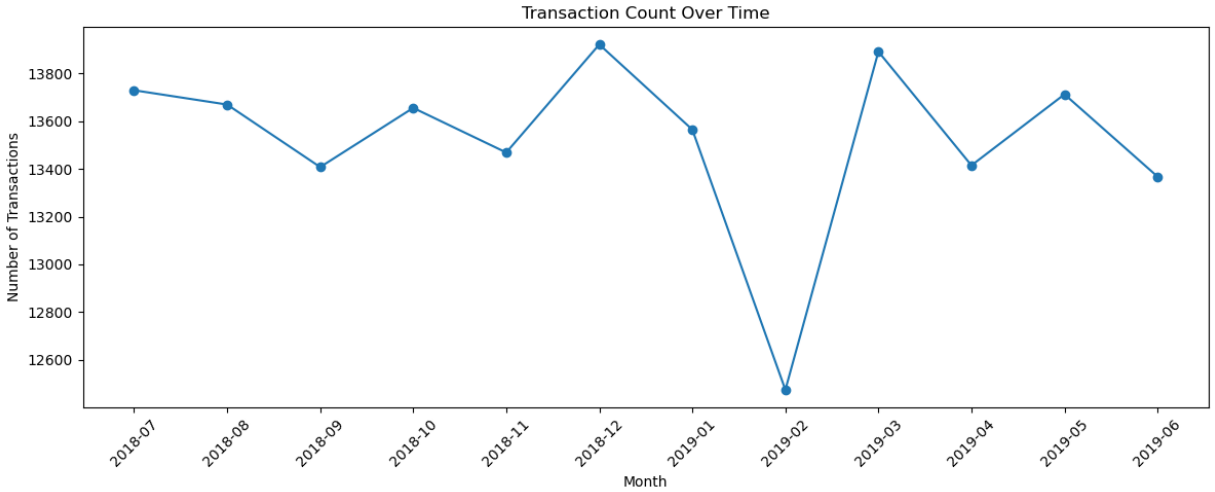
Out[138…
```
DATE              0
STORE_NBR         0
LYLTY_CARD_NBR    0
TXN_ID            0
PROD_NBR          0
PROD_NAME         0
PROD_QTY          0
TOT_SALES         0
LIFESTAGE         0
PREMIUM_CUSTOMER  0
dtype: int64
```

In [139…
```
import matplotlib.pyplot as plt
```

In [140…
```
concat_data['DATE'] = pd.to_datetime(concat_data['DATE'])
concat_data = concat_data.sort_values('DATE')

monthly_counts = concat_data.groupby(concat_data['DATE'].dt.to_period('M')).size().
monthly_counts['DATE'] = monthly_counts['DATE'].astype(str)

plt.figure(figsize=(12, 5))
plt.plot(monthly_counts['DATE'], monthly_counts['count'], marker='o')
plt.title('Transaction Count Over Time')
plt.xlabel('Month')
plt.ylabel('Number of Transactions')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



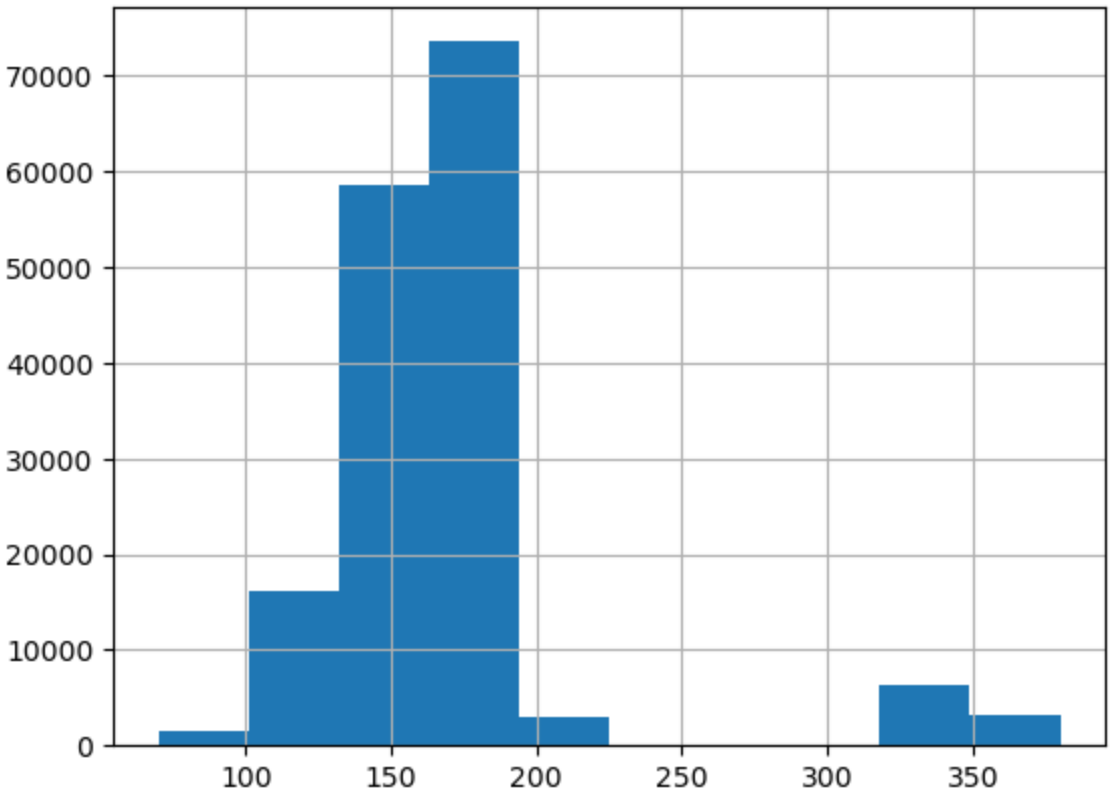
PACK SIZE

In [141…
```
concat_data['PACK_SIZE'] = concat_data['PROD_NAME'].str.extract(r'(\d+)').astype(in
```

In [142…
```
print(f"Max pack size: {concat_data['PACK_SIZE'].max()} \nMin pack size: {concat_da
```

```
Max pack size: 380
Min pack size: 70
```

In [143…
```
concat_data['PACK_SIZE'].hist()
plt.show()
```



BRANDS

In [144…
```
concat_data['BRAND'] = concat_data['PROD_NAME'].str.split().str[0]
```

In [152…
```
concat_data.head()
```

Out[152…

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PR
151499	2018-07-01	113	113012	115430	103	RRD Steak & Chimuchurri 150g	
46995	2018-07-01	156	156011	156670	14	Smiths Crnkle Chip Orgnl Big Bag 380g	
68339	2018-07-01	165	165354	167050	63	Kettle 135g Swt Pot Sea Salt	
20417	2018-07-01	10	10055	9352	36	Kettle Chilli 175g	
20435	2018-07-01	10	10111	9693	82	Smith Crinkle Cut Mac N Cheese 150g	

In [153…
```
concat_data['BRAND'].unique()
```

Out[153…
```
array(['Red Rock Deli', 'Smiths', 'Kettle', 'Woolworths', 'Thins',
       'Tyrrells', 'Pringles', 'Infuzions', 'Cobs Popcorn',
       'Natural Chip Co', 'French Fries', 'Nice & Natural',
       'Burger Rings'], dtype=object)
```

In [149…
```
brand_mapping = {
    'NCC'       : 'Nice & Natural',
    'Thins'     : 'Thins',
    'WW'        : 'Woolworths',
    'Smiths'    : 'Smiths',
    'Kettle'    : 'Kettle',
    'French'    : 'French Fries',
    'Cobs'      : 'Cobs Popcorn',
    'Infuzions' : 'Infuzions',
    'RRD'       : 'Red Rock Deli',
    'Pringles'  : 'Pringles',
    'Infzns'    : 'Infuzions',       # likely a typo/duplicate
    'Tyrrells'  : 'Tyrrells',
    'Smith'     : 'Smiths',          # likely a typo/duplicate
    'Red'       : 'Red Rock Deli',   # likely part of Red Rock Deli
    'Burger'    : 'Burger Rings',
    'Natural'   : 'Natural Chip Co',
    'Snbts'     : 'Sunbites'
}

# Apply it
concat_data['BRAND'] = concat_data['BRAND'].map(brand_mapping)
```

In [154…
```
concat_data.head()
```

Out[154…

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PR
151499	2018-07-01	113	113012	115430	103	RRD Steak & Chimuchurri 150g	
46995	2018-07-01	156	156011	156670	14	Smiths Crnkle Chip Orgnl Big Bag 380g	
68339	2018-07-01	165	165354	167050	63	Kettle 135g Swt Pot Sea Salt	
20417	2018-07-01	10	10055	9352	36	Kettle Chilli 175g	
20435	2018-07-01	10	10111	9693	82	Smith Crinkle Cut Mac N Cheese 150g	

In [172…
```
concat_data['PRICE'] = concat_data['TOT_SALES'] / concat_data['PROD_QTY']
concat_data.head()
```

Out[172…

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PR
151499	2018-07-01	113	113012	115430	103	RRD Steak & Chimuchurri 150g	
46995	2018-07-01	156	156011	156670	14	Smiths Crnkle Chip Orgnl Big Bag 380g	
68339	2018-07-01	165	165354	167050	63	Kettle 135g Swt Pot Sea Salt	
20417	2018-07-01	10	10055	9352	36	Kettle Chilli 175g	
20435	2018-07-01	10	10111	9693	82	Smith Crinkle Cut Mac N Cheese 150g	

Distribution of customers by each group

In [163…
```
# Life Stage Ranked and Count of Customers in each group
concat_data.groupby('LIFESTAGE').size().reset_index(name = 'count').sort_values('co
# Older Singles/couples spend the most on potato chips
```

Out[163…

	LIFESTAGE	count
3	OLDER SINGLES/COUPLES	33206
2	OLDER FAMILIES	30300
4	RETIREES	30252
5	YOUNG FAMILIES	26880
6	YOUNG SINGLES/COUPLES	22078
0	MIDAGE SINGLES/COUPLES	15359
1	NEW FAMILIES	4207

In [164…

```
# Premium Ranked and count of customers in each group
concat_data.groupby('PREMIUM_CUSTOMER').size().reset_index(name = 'count').sort_val
# Mainstream spends the most on potato chips
```

Out[164…

	PREMIUM_CUSTOMER	count
1	Mainstream	61982
0	Budget	57323
2	Premium	42977

In [165…

```
# Combined ranked, Premium, and Life Stage
concat_data.groupby(['LIFESTAGE','PREMIUM_CUSTOMER']).size().reset_index(name = 'co
```

Out[165…

	LIFESTAGE	PREMIUM_CUSTOMER	count
6	OLDER FAMILIES	Budget	14426
13	RETIREES	Mainstream	13041
19	YOUNG SINGLES/COUPLES	Mainstream	12362
15	YOUNG FAMILIES	Budget	11701
9	OLDER SINGLES/COUPLES	Budget	11254
10	OLDER SINGLES/COUPLES	Mainstream	11181
11	OLDER SINGLES/COUPLES	Premium	10771
12	RETIREES	Budget	9211
7	OLDER FAMILIES	Mainstream	8865
14	RETIREES	Premium	8000
16	YOUNG FAMILIES	Mainstream	7965
17	YOUNG FAMILIES	Premium	7214
1	MIDAGE SINGLES/COUPLES	Mainstream	7158
8	OLDER FAMILIES	Premium	7009
18	YOUNG SINGLES/COUPLES	Budget	5770
2	MIDAGE SINGLES/COUPLES	Premium	5084
20	YOUNG SINGLES/COUPLES	Premium	3946
0	MIDAGE SINGLES/COUPLES	Budget	3117
3	NEW FAMILIES	Budget	1844
4	NEW FAMILIES	Mainstream	1410
5	NEW FAMILIES	Premium	953

Distribution of Sales by each group

In [166…

```
# Combined Group
concat_data.groupby(['LIFESTAGE', 'PREMIUM_CUSTOMER'])['TOT_SALES'].sum().reset_ind
```

Out[166…

	LIFESTAGE	PREMIUM_CUSTOMER	TOT_SALES
6	OLDER FAMILIES	Budget	101776.1
13	RETIREES	Mainstream	91649.1
19	YOUNG SINGLES/COUPLES	Mainstream	90420.9
15	YOUNG FAMILIES	Budget	82752.3
9	OLDER SINGLES/COUPLES	Budget	81277.1
10	OLDER SINGLES/COUPLES	Mainstream	79093.6
11	OLDER SINGLES/COUPLES	Premium	77735.2
12	RETIREES	Budget	66493.3
7	OLDER FAMILIES	Mainstream	62559.2
14	RETIREES	Premium	57967.7
16	YOUNG FAMILIES	Mainstream	55840.7
1	MIDAGE SINGLES/COUPLES	Mainstream	53095.1
17	YOUNG FAMILIES	Premium	51029.6
8	OLDER FAMILIES	Premium	49192.5
18	YOUNG SINGLES/COUPLES	Budget	37091.5
2	MIDAGE SINGLES/COUPLES	Premium	35189.5
20	YOUNG SINGLES/COUPLES	Premium	25550.8
0	MIDAGE SINGLES/COUPLES	Budget	21309.2
3	NEW FAMILIES	Budget	13041.8
4	NEW FAMILIES	Mainstream	9856.1
5	NEW FAMILIES	Premium	6775.2

In [167…

```
# Member Status Only
concat_data.groupby('PREMIUM_CUSTOMER')['TOT_SALES'].sum().reset_index().sort_value
```

Out[167…

	PREMIUM_CUSTOMER	TOT_SALES
1	Mainstream	442514.7
0	Budget	403741.3
2	Premium	303440.5

In [168…

```
# Lifestage only
concat_data.groupby('LIFESTAGE')['TOT_SALES'].sum().reset_index().sort_values('TOT_
```

Out[168…

	LIFESTAGE	TOT_SALES
3	OLDER SINGLES/COUPLES	238105.9
4	RETIREES	216110.1
2	OLDER FAMILIES	213527.8
5	YOUNG FAMILIES	189622.6
6	YOUNG SINGLES/COUPLES	153063.2
0	MIDAGE SINGLES/COUPLES	109593.8
1	NEW FAMILIES	29673.1

Quantity of Chips

In [169…

```
# Combined Group
concat_data.groupby(['LIFESTAGE', 'PREMIUM_CUSTOMER'])['PROD_QTY'].sum().reset_inde
```

Out[169…

	LIFESTAGE	PREMIUM_CUSTOMER	PROD_QTY
6	OLDER FAMILIES	Budget	28052
13	RETIREES	Mainstream	24560
19	YOUNG SINGLES/COUPLES	Mainstream	22866
15	YOUNG FAMILIES	Budget	22721
9	OLDER SINGLES/COUPLES	Budget	21541
10	OLDER SINGLES/COUPLES	Mainstream	21341
11	OLDER SINGLES/COUPLES	Premium	20601
12	RETIREES	Budget	17421
7	OLDER FAMILIES	Mainstream	17272
16	YOUNG FAMILIES	Mainstream	15468
14	RETIREES	Premium	15223
17	YOUNG FAMILIES	Premium	13967
1	MIDAGE SINGLES/COUPLES	Mainstream	13680
8	OLDER FAMILIES	Premium	13642
18	YOUNG SINGLES/COUPLES	Budget	10364
2	MIDAGE SINGLES/COUPLES	Premium	9602
20	YOUNG SINGLES/COUPLES	Premium	7099
0	MIDAGE SINGLES/COUPLES	Budget	5871
3	NEW FAMILIES	Budget	3418
4	NEW FAMILIES	Mainstream	2599
5	NEW FAMILIES	Premium	1770

In [170…

```
# Member Type
concat_data.groupby('PREMIUM_CUSTOMER')['PROD_QTY'].sum().reset_index().sort_values
```

Out[170…

	PREMIUM_CUSTOMER	PROD_QTY
1	Mainstream	117786
0	Budget	109388
2	Premium	81904

In [171…

```
# Lifestage
concat_data.groupby('LIFESTAGE')['PROD_QTY'].sum().reset_index().sort_values('PROD_
```

Out[171…

	LIFESTAGE	PROD_QTY
3	OLDER SINGLES/COUPLES	63483
2	OLDER FAMILIES	58966
4	RETIREES	57204
5	YOUNG FAMILIES	52156
6	YOUNG SINGLES/COUPLES	40329
0	MIDAGE SINGLES/COUPLES	29153
1	NEW FAMILIES	7787

Average Chip Price

In [174…

```
# Combined
concat_data.groupby(['LIFESTAGE', 'PREMIUM_CUSTOMER'])['PRICE'].mean().reset_index(
```

Out[174...

	LIFESTAGE	PREMIUM_CUSTOMER	PRICE
19	YOUNG SINGLES/COUPLES	Mainstream	3.946174
1	MIDAGE SINGLES/COUPLES	Mainstream	3.880148
5	NEW FAMILIES	Premium	3.813116
3	NEW FAMILIES	Budget	3.810033
12	RETIREES	Budget	3.808886
14	RETIREES	Premium	3.803925
11	OLDER SINGLES/COUPLES	Premium	3.768248
4	NEW FAMILIES	Mainstream	3.768085
9	OLDER SINGLES/COUPLES	Budget	3.767318
13	RETIREES	Mainstream	3.723541
10	OLDER SINGLES/COUPLES	Mainstream	3.700295
17	YOUNG FAMILIES	Premium	3.656277
2	MIDAGE SINGLES/COUPLES	Premium	3.654013
15	YOUNG FAMILIES	Budget	3.641793
6	OLDER FAMILIES	Budget	3.627180
7	OLDER FAMILIES	Mainstream	3.623982
0	MIDAGE SINGLES/COUPLES	Budget	3.623276
16	YOUNG FAMILIES	Mainstream	3.612185
8	OLDER FAMILIES	Premium	3.605507
20	YOUNG SINGLES/COUPLES	Premium	3.574911
18	YOUNG SINGLES/COUPLES	Budget	3.553778

In [175...

```
# membership
concat_data.groupby('PREMIUM_CUSTOMER')['PRICE'].mean().reset_index().sort_values('
```

Out[175...

	PREMIUM_CUSTOMER	PRICE
1	Mainstream	3.754300
2	Premium	3.699283
0	Budget	3.685155

In [176...

```
# Lifestage
concat_data.groupby('LIFESTAGE')['PRICE'].mean().reset_index().sort_values('PRICE',
```

Out[176…

	LIFESTAGE	PRICE
1	NEW FAMILIES	3.796672
6	YOUNG SINGLES/COUPLES	3.777267
4	RETIREES	3.770784
0	MIDAGE SINGLES/COUPLES	3.753164
3	OLDER SINGLES/COUPLES	3.745052
5	YOUNG FAMILIES	3.636907
2	OLDER FAMILIES	3.621231

Mainstream midage and young singles and couples are more willing to pay more per packet of chips compared to their budget and premium counterparts. This may be due to premium shoppers being more likely to buy healthy snacks and when they buy chips, this is mainly for entertainment purposes rather than their own consumption. This is also supported by there being fewer premium midage and young singles and couples buying chips compared to their mainstream counterparts.

In [177…

```
concat_data[concat_data['PREMIUM_CUSTOMER'] == 'Premium'].groupby(['LIFESTAGE','PRE
```

Out[177…

	LIFESTAGE	PREMIUM_CUSTOMER	count
3	OLDER SINGLES/COUPLES	Premium	10771
4	RETIREES	Premium	8000
5	YOUNG FAMILIES	Premium	7214
2	OLDER FAMILIES	Premium	7009
0	MIDAGE SINGLES/COUPLES	Premium	5084
6	YOUNG SINGLES/COUPLES	Premium	3946
1	NEW FAMILIES	Premium	953

As the difference in average price per unit isn't large, we can check if this difference is statistically different.

In [178…

```
from scipy import stats

# Filter groups
mainstream_midage = concat_data[
    (concat_data['PREMIUM_CUSTOMER'] == 'Mainstream') &
    (concat_data['LIFESTAGE'] == 'MIDAGE SINGLES/COUPLES')
]['PRICE']

budget_premium_midage = concat_data[
    (concat_data['PREMIUM_CUSTOMER'].isin(['Premium', 'Budget'])) &
```

```
    (concat_data['LIFESTAGE'] == 'MIDAGE SINGLES/COUPLES')
]['PRICE']

# T-test for Midage Singles/Couples
t_stat, p_value = stats.ttest_ind(mainstream_midage, budget_premium_midage)
print(f"Midage Singles/Couples — t-stat: {t_stat:.4f}, p-value: {p_value:.4f}")


# Filter groups
mainstream_young = concat_data[
    (concat_data['PREMIUM_CUSTOMER'] == 'Mainstream') &
    (concat_data['LIFESTAGE'] == 'YOUNG SINGLES/COUPLES')
]['PRICE']

budget_premium_young = concat_data[
    (concat_data['PREMIUM_CUSTOMER'].isin(['Premium', 'Budget'])) &
    (concat_data['LIFESTAGE'] == 'YOUNG SINGLES/COUPLES')
]['PRICE']

# T-test for Young Singles/Couples
t_stat, p_value = stats.ttest_ind(mainstream_young, budget_premium_young)
print(f"Young Singles/Couples — t-stat: {t_stat:.4f}, p-value: {p_value:.4f}")
```

```
Midage Singles/Couples — t-stat: 14.2035, p-value: 0.0000
Young Singles/Couples — t-stat: 27.8544, p-value: 0.0000
```

In [179…

```
print("=== Midage Singles/Couples ===")
print(f"Mainstream mean price:      {mainstream_midage.mean():.4f}")
print(f"Budget/Premium mean price:  {budget_premium_midage.mean():.4f}")

print("\n=== Young Singles/Couples ===")
print(f"Mainstream mean price:      {mainstream_young.mean():.4f}")
print(f"Budget/Premium mean price:  {budget_premium_young.mean():.4f}")
```

```
=== Midage Singles/Couples ===
Mainstream mean price:      3.8801
Budget/Premium mean price:  3.6423

=== Young Singles/Couples ===
Mainstream mean price:      3.9462
Budget/Premium mean price:  3.5624
```

The t-test results in a p-value < 0.01, i.e., the unit price for mainstream young and mid-aged singles and couples is significantly higher than that of budget or premium, young and middle-aged singles and couples.

Deep Drive into segments: Older Families Budget

In [180…

```
segment_1 = concat_data[(concat_data['PREMIUM_CUSTOMER'] == 'Budget')&(concat_data[
segment_1.head()
```

Out[180…

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PRC
20417	2018-07-01	10	10055	9352	36	Kettle Chilli 175g	
20435	2018-07-01	10	10111	9693	82	Smith Crinkle Cut Mac N Cheese 150g	
31498	2018-07-01	219	219154	218894	43	Smith Crinkle Cut Bolognese 150g	
20234	2018-07-01	5	5085	4863	11	RRD Pc Sea Salt 165g	
21682	2018-07-01	39	39224	36000	37	Smiths Thinly Swt Chli&S/ Cream175G	

In [183…

```
# Which brand is most purchased in the most transactions
segment_1.groupby('BRAND').size().reset_index(name = 'count').sort_values('count',a
```

Out[183…

	BRAND	count
9	Smiths	3093
4	Kettle	2563
7	Pringles	1996
8	Red Rock Deli	1857
10	Thins	1171
12	Woolworths	773
1	Cobs Popcorn	760
3	Infuzions	682
5	Natural Chip Co	576
11	Tyrrells	489
6	Nice & Natural	165
0	Burger Rings	159
2	French Fries	142

In [192…

```
# Which brand sold the most units
segment_1.groupby('BRAND')['PROD_QTY'].sum().reset_index().sort_values('PROD_QTY',
```

Out[192...

	BRAND	PROD_QTY
9	Smiths	5999
4	Kettle	5008
7	Pringles	3865
8	Red Rock Deli	3604
10	Thins	2278
12	Woolworths	1510
1	Cobs Popcorn	1489
3	Infuzions	1328
5	Natural Chip Co	1122
11	Tyrrells	953
6	Nice & Natural	319
0	Burger Rings	301
2	French Fries	276

In [184...

```
# Which brand is the most profitable for this segment
segment_1.groupby('BRAND')['TOT_SALES'].sum().reset_index().sort_values('TOT_SALES'
```

Out[184...

	BRAND	TOT_SALES
4	Kettle	25231.6
9	Smiths	21651.1
7	Pringles	14300.5
8	Red Rock Deli	10162.2
10	Thins	7517.4
1	Cobs Popcorn	5658.2
3	Infuzions	4660.0
11	Tyrrells	4002.6
5	Natural Chip Co	3366.0
12	Woolworths	2749.2
6	Nice & Natural	957.0
2	French Fries	828.0
0	Burger Rings	692.3

In [186…
```
segment_1.groupby('BRAND')['PRICE'].mean().reset_index().sort_values('PRICE', ascen
```

Out[186…

	BRAND	PRICE
4	Kettle	5.037924
11	Tyrrells	4.200000
1	Cobs Popcorn	3.800000
7	Pringles	3.700000
9	Smiths	3.611219
3	Infuzions	3.508504
10	Thins	3.300000
2	French Fries	3.000000
5	Natural Chip Co	3.000000
6	Nice & Natural	3.000000
8	Red Rock Deli	2.820194
0	Burger Rings	2.300000
12	Woolworths	1.821604

In [195…
```
segment_1.groupby('PACK_SIZE')['TOT_SALES'].sum().reset_index().sort_values('TOT_SA
```

Out[195…

	PACK_SIZE	TOT_SALES
8	175	34346.5
4	150	16403.8
2	134	14300.5
1	110	9655.8
6	165	8316.6
11	330	6150.3
7	170	4312.3
12	380	2991.3
3	135	2213.4
5	160	1128.6
10	220	692.3
0	70	662.4
9	200	602.3

Older Families Budget Insight: Although Smith's sold the most to Budget Older Families, Kettle generated the most revenue. Despite lower sales volume, Kettle's higher revenue suggests stronger brand value per unit, indicating a customer segment willing to pay a premium. This higher price point, however, appears to suppress overall demand. A controlled price promotion on Kettle would allow the business to measure price elasticity and determine whether reducing price drives enough volume to offset the margin reduction, ultimately helping to identify whether price or preference is the primary driver of Kettle's lower unit sales. Furthermore, 175-gram size bags generate the most profit.

Deep dive into segments: Mainstream, young singles/couples

In [187…
```
segment_2 = concat_data[(concat_data['PREMIUM_CUSTOMER'] == 'Mainstream')&(concat_d
segment_2.head()
```

Out[187…

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PR
151499	2018-07-01	113	113012	115430	103	RRD Steak & Chimuchurri 150g	
151667	2018-07-01	116	116179	120232	81	Pringles Original Crisps 134g	
151329	2018-07-01	109	109063	110536	98	NCC Sour Cream & Garden Chives 175g	
149211	2018-07-01	59	59307	55832	17	Kettle Sensations BBQ&Maple 150g	
149279	2018-07-01	61	61145	57698	11	RRD Pc Sea Salt 165g	

In [188…
```
# Which brand is purchased during the most transactions
segment_2.groupby('BRAND').size().reset_index(name = 'count').sort_values('count',a
```

Out[188...

	BRAND	count
4	Kettle	2949
7	Pringles	2315
9	Smiths	1988
10	Thins	1166
8	Red Rock Deli	969
1	Cobs Popcorn	864
3	Infuzions	657
11	Tyrrells	619
5	Natural Chip Co	321
12	Woolworths	301
2	French Fries	78
6	Nice & Natural	73
0	Burger Rings	62

In [193...

```
# Which brand sold the most units
segment_2.groupby('BRAND')['PROD_QTY'].sum().reset_index().sort_values('PROD_QTY',
```

Out[193...

	BRAND	PROD_QTY
4	Kettle	5497
7	Pringles	4326
9	Smiths	3609
10	Thins	2187
8	Red Rock Deli	1753
1	Cobs Popcorn	1617
3	Infuzions	1227
11	Tyrrells	1143
5	Natural Chip Co	578
12	Woolworths	548
2	French Fries	143
6	Nice & Natural	132
0	Burger Rings	106

In [190…
```
# Which brand is the most profitable for this segment
segment_2.groupby('BRAND')['TOT_SALES'].sum().reset_index().sort_values('TOT_SALES'
```

Out[190…

	BRAND	TOT_SALES
4	Kettle	27718.6
7	Pringles	16006.2
9	Smiths	15265.7
10	Thins	7217.1
1	Cobs Popcorn	6144.6
8	Red Rock Deli	4958.1
11	Tyrrells	4800.6
3	Infuzions	4508.6
5	Natural Chip Co	1734.0
12	Woolworths	998.6
2	French Fries	429.0
6	Nice & Natural	396.0
0	Burger Rings	243.8

In [191…
```
segment_2.groupby('BRAND')['PRICE'].mean().reset_index().sort_values('PRICE', ascen
```

Out[191…

	BRAND	PRICE
4	Kettle	5.042455
9	Smiths	4.205936
11	Tyrrells	4.200000
1	Cobs Popcorn	3.800000
7	Pringles	3.700000
3	Infuzions	3.665753
10	Thins	3.300000
2	French Fries	3.000000
5	Natural Chip Co	3.000000
6	Nice & Natural	3.000000
8	Red Rock Deli	2.826625
0	Burger Rings	2.300000
12	Woolworths	1.822924

In [196…
```
segment_2.groupby('PACK_SIZE')['TOT_SALES'].sum().reset_index().sort_values('TOT_SA
```

Out[196…

	PACK_SIZE	TOT_SALES
8	175	29879.4
2	134	16006.2
4	150	11912.9
1	110	10389.2
6	165	7050.6
11	330	6378.3
12	380	3516.4
3	135	2247.0
7	170	1896.6
5	160	440.8
0	70	264.0
10	220	243.8
9	200	195.7

Mainstream Young Singles/Couples Insight: Kettle led across all key metrics; unit sales, total revenue. Price was consistent with findings across other segments. Unlike Budget Older Families, who showed sensitivity to Kettle's premium pricing, Mainstream Young Singles/ Couples appear less price sensitive, willing to pay a premium for Kettle despite cheaper alternatives like Smiths being available. This is further supported by Kettle generating more revenue from Mainstream Young Singles/Couples than from Budget Older Families, suggesting that disposable income and lifestyle play a significant role in purchasing decisions. Notably, Pringles also ranked above Smiths in this segment, reinforcing the pattern that Mainstream Young Singles/Couples gravitate toward premium or well-known brands over value options. Pack size remained consistent with both segments.

In [197…
```
concat_data.to_csv('ChipDataCleaned.csv', index=False, encoding='utf-8-sig', sep=',
```